Predicting Age from the Transcriptome of Human Dermal Fibroblasts

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ABSTRACT

Accurately estimating biological age remains a fundamental challenge in aging research, with implications for disease prediction and health interventions. RNA sequencing (RNA-seq) data has emerged as a promising direction by capturing dynamic gene expression changes driven by biological and environmental factors. Building on the work of the original analysis [1], which demonstrated that machine learning models trained on RNA-seq transcriptome data from human dermal fibroblasts could predict chronological age with a high degree of accuracy, this study seeks to reproduce those findings and extend them through the implementation of more advanced modeling frameworks. Beyond predictive performance, we conduct deeper analyses of the underlying transcriptomic data to identify genetic pathways associated with aging. By combining additional modeling techniques and data analysis, this work aims to better understand the mechanisms driving aging and contribute to the broader goal of translating biological age estimation.

INTRODUCTION

A major goal in aging research is to identify biomarkers that can estimate biological age more accurately than chronological age. In the original analysis, researchers sought to develop a machine learning method to predict chronological age using RNA-seq transcriptome data from human dermal fibroblasts [1]. Testing a wide range of models, from simple linear regression to ensemble-based models, the researchers were able to generate results that outperform previous studies on age prediction. In the proposed analysis, we hope to continue to expand on these results. The researchers mainly focused on developing the best model, whereas our study looks to reproduce the initial findings and develop additional, more complex models like XGBoost and Neural Networks to further increase accuracy. Additionally, we want to dive deeper into the data to explore the genetic pathways that affect this.

Earlier approaches relied heavily on physiological indicators (hearing/vision, lung capacity); however, this fails to incorporate lifestyle and organ health factors. The goal in the field has also transitioned from merely predicting biological age to predicting future disease risk and overall health outcomes. More recently, molecular biomarkers, specifically DNA methylation, have become the foundation of understanding our "epigenetic clocks" [2], with highly accurate models capable of predicting age. At the same time, other approaches using RNA-seq data have gained popularity, as gene expression captures dynamic biological processes and environmental influences on aging. Moreover, recent studies have further expanded these methods by leveraging large-scale, multi-tissue datasets and more advanced machine learning frameworks, such as deep learning and mixed models, to better capture tissue-specific aging patterns [5]. Thus, this topic is extremely important, as it not only helps us understand our personal health but also enhances our ability to detect early signs of disease and provide timely interventions.

METHODOLOGY

Dataset and Preprocessing

The dataset used in the analysis was obtained from the Coriell Institute cell repository and the Progeria Research Foundation. The samples were derived from 133 "apparently healthy individuals" whose ages range from 1 to 94 years, and an additional 10 patients with Hutchinson-Gilford progeria syndrome (HGPS), a premature aging disease. The structure of the dataset follows where columns represent individual genetic transcripts (27,142 original entries) and rows represent diverse biological samples (143 total individuals). The first eight features contain descriptive metadata for each gene, including the Transcript ID, chromosomal location, and length, while the subsequent columns contain the FPKM (Fragments Per Kilobase of transcript per Million mapped reads) values, which quantify gene expression levels.

To prepare this data for predictive modeling, we implemented a rigorous cleaning pipeline that mirrored the original research code. First, the genomic metadata in the first eight columns were isolated and included in a new metadata-type data frame. We then applied a rigorous feature selection process. Rather than applying a simple zero-variance filter, we excluded any transcript that failed to meet a minimum expression threshold (Max FPKM > 5.0) or lacked a significant fold change (Max/Min > 5.0), ensuring the model trained only on highly informative and variable genes. We then applied a log2 transformation to scale the gene expression values, which helped stabilize the data by reducing extreme skewness in the raw values and preventing Principal Component Analysis (PCA) from being dominated by outliers. Using the Tracking IDs from the sample headers to join the genomic data with a secondary metadata sheet, we extracted labels such as age and sex from long descriptive text strings (for example, 101_19yr_Female_Caucasian), producing a structured dataset ready for supervised learning, as seen in Table 1. Additionally, there was no set way we scaled the values; the methods varied per model.

	Tracking_ID	Age_Numeric	Gender	Condition	NM_006918	NM_001145318
101_19yr_Female_Caucasian/ FPKM	101	19	female	normal	2.615181	3.909869
102_19yr_Male_Caucasian/ FPKM	102	19	male	normal	2.232968	3.833396
103_19yr_Male_Caucasian/ FPKM	103	19	male	normal	2.123004	3.394514
104_19yr_Male_Caucasian/ FPKM	104	19	male	normal	1.788686	3.444137
105_20yr_Male_Caucasian/ FPKM	105	20	male	normal	2.753818	3.820792
106_20yr_Female_Caucasian/ FPKM	106	20	female	normal	2.544485	3.227125
107_31yr_Female_Caucasian/ FPKM	107	31	female	normal	2.291309	3.856388
108_31_female_Caucasian/ FPKM	108	31	female	normal	2.721154	4.054501
109_32_male_Unknown/ FPKM	109	32	male	normal	3.280808	4.252855
110_32_female_BlackPuertoRican/ FPKM	110	32	female	normal	2.669254	3.923720

Table 1. Sample metadata and gene expression dataset preview.

Dimensionality Reduction

A gene correlation heatmap in Figure 1 was generated to identify the degree of collinearity across the feature set. The heatmap revealed large clusters of both high positive and negative correlations across the top 500 genes, providing strong justification for dimensionality reduction. Given these structured correlation patterns, PCA [3] was performed on the refined feature subset to extract the most dominant sources of variation in the genomic data.

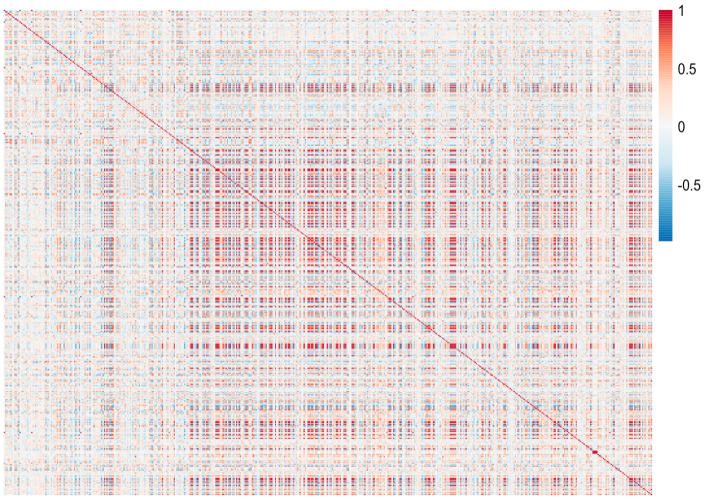


Figure 1. Correlation matrix of the top 500 genes showing high correlations that highlight redundancy in transcriptomic features.

The results of the PCA indicated that a substantial proportion of the variance was explained by the first two principal components (~40%), as supported by the elbow method, which identified two components as the optimal point before the curve begins to flatten (Figure 2).

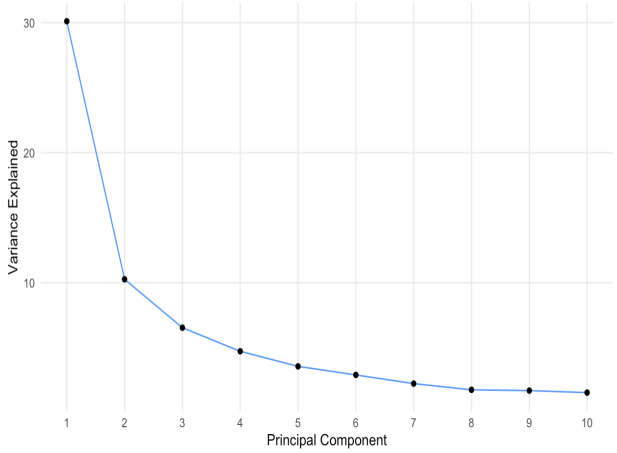


Figure 2. Scree plot illustrating the variance explained by each principal component to determine the optimal number of components.

The PCA biplot in Figure 3 is a diagnostic graph used to identify clusters, trends, and outliers, and in this case to verify underlying aging patterns between our healthy and HGPS groups. The plot shows underlying structure in the data, where samples progress from left to right and the color gradient shifts from dark purple to yellow, indicating a predictable linear age-related trajectory from younger to older samples. In addition, this mapping provides evidence of accelerated biological aging in the HGPS samples, represented as triangles. Despite being chronologically young, these samples are significantly displaced along PC1, clustering with healthy individuals in their 40s and 50s rather than with age-matched peers. Overall, this visualization supports the presence of distinct molecular aging signatures when comparing the HGPS group to the healthy baseline.

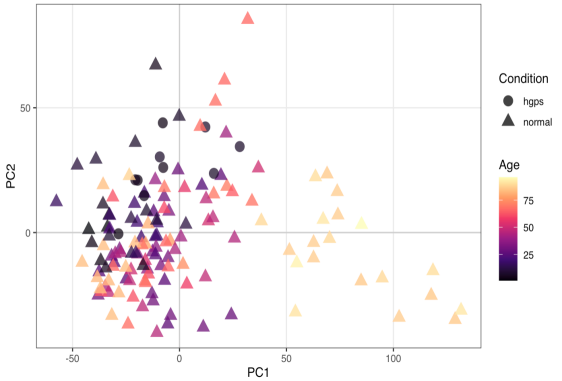


Figure 3. PCA cluster plot showing separation by health condition along the aging trajectory.

Exploratory Analysis

Further using the findings from the PCA model, we conducted a preliminary analysis to explore the genetic pathways associated with aging [6]. While the original study focused primarily on building machine learning models for age prediction, we included this step to dive deeper into the biological processes that cause aging, as well as deepen our understanding of the topic to better contextualize patterns observed in the models. To analyze this, we visualized the genes most influential under PC1, as shown in Figure 4. We focused solely on PC1, given that it alone explained over 30% of the variance in the dataset. The plot reveals two distinct genetic signatures that define aging. Genes with high positive weight on PC1 are primarily associated with extracellular matrix organization and skin structure, led by the gene NM_003249. In contrast, genes with the strongest negative weights, such as NM_001810, are linked to cell cycle regulation and mitosis. Briefly mentioning code syntax, NM prefixes denote protein-coding messenger RNA (mRNA) transcripts, while NR denotes non-coding RNA transcripts (which were also impactful but not shown in this limited visual). Overall, this supports the main theme of the study by linking aging to the molecular breakdown of skin structure, particularly through structural protein-related pathways.

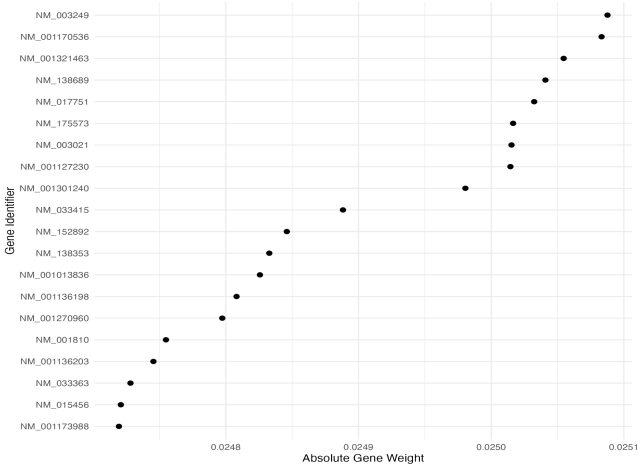


Figure 4. Dominant PC1 genes highlight pathways associated with underlying aging.

Cross-Validation Strategy

After performing PCA, the dataset was split for modeling. Rather than replicating the original study's 10-fold cross-validation, we employed leave-one-out cross-validation (LOOCV), given the small sample size and the prior dimensionality reduction step. In LOOCV, the model is trained N times using all but one individual for training and testing on that single omitted sample. This approach was chosen to maximize the use of available training data while preventing overfitting.

Model Development

The first recreated model from the original study was the Ensemble LDA [4]. This was the paper's best performing model and worked by repeating classification tasks across overlapping age bins rather than direct regression. The algorithm divides the age range into bins of width class_size = 20 years, repeats this partition with offsets of 0 through 19, and trains a separate LDA classifier for each partition. Each classifier votes for all integer ages within its predicted bin, and the final predicted age is the mode across all votes. In the original Python implementation, class_size was grid-searched over values of 10, 20, and 30, though this search was commented out in the available notebook, indicating that class_size = 20 had already been identified as the optimal value.

The second recreated model was Elastic Net regression which combines L1 and L2 regularization penalties, enabling simultaneous variable selection and coefficient shrinkage. The original study tuned the L1 mixing ratio and regularization strength via grid search and repeated 10-fold cross validation but loaded a pre-saved best model rather than rerunning the search, meaning the exact optimal hyperparameters are not recoverable. Our reproduction searched the same grid with a set seed for reproducibility. The grid covered alpha (L1 mixing ratio) from 0.0 to 1.0 in steps of 0.1, and lambda (regularization strength) from 0.1 to 1.0 in steps of 0.1, yielding 110 combinations each evaluated across repeated 10-fold cross validation. The best parameters identified were a mix of L1 mixing ratio of 0.1, and a regularization strength of 1.00.

The first new model was a Neural Network structured with an input layer, a single hidden layer containing five neurons, and an output neuron for age prediction. Z-score scaling was applied to the PCA components to prevent high-variance components from disproportionately influencing network weights. The development process was hindered by several challenges, most notably the failure of the model to converge during initial training runs. Because biological data is inherently noisy, the algorithm would stall or reach a local minimum, resulting in errors where the model could not find a solution. This was addressed by significantly increasing the stepmax to 1,000,000 to allow the network more iterations to reach a solution and loosening the convergence threshold to 0.1. By limiting the architecture to one hidden layer, we successfully captured non-linear trends while avoiding the over-parameterization that often leads to overfitting in smaller datasets.

The second new model was an eXtreme Gradient Boosting (XGBoost) model, a tree-based ensemble method designed to capture non-linear patterns that simpler models may miss. We experimented with various hyperparameters, testing variations in tree depth and learning rates. The model was tested at max_depth values of 3-5, learning rates of .05, .01, and .02, and a gamma of 0 and 1. The most optimal parameters ended up being a depth of 3, gamma of 0, and a learning rate of .05.

RESULTS

Reproduced Models

The Ensemble LDA model produced the strongest results among the reproduced models, achieving an MAE of 12.1 years, a MedAE of 5.0 years, and an R^2 of 0.60 (Table 2). Overall, this replicates the main conclusion of the original study, where the ensemble approach

outperformed all other evaluated models. The MedAE remains relatively close to the original reported value, suggesting that the model is successfully capturing the central tendency of age prediction. However, the higher MAE and lower R^2 indicate reduced performance in handling larger prediction errors and overall variance explained. These differences are consistent with methodological variation in the LDA implementation and regularization approach. In particular, differences in shrinkage estimation and default parameterization across programming language implementations of LDA may affect sensitivity to extreme predictions, which likely contributes to the observed increase in MAE. The relatively stable MedAE suggests that performance degradation is driven primarily by outlier behavior rather than a failure to capture the underlying signal.

LDA	MAE	MedAE	R^2
Original	7.7	4.0	0.81
Reproduced	12.1	5.0	0.60

Table 2. Performance metrics for Ensemble LDA in the original and reproduced experiments.

The Elastic Net regression achieved an MAE of 14.3 years, a MedAE of 12.4 years, and an R^2 of 0.63, as shown in Table 3. Although the metrics are closely similar, the reproduced model ultimately falls short of the original performance. The most likely explanation is hyperparameter sensitivity. The original notebook loaded a pre-saved best model rather than rerunning the grid search, meaning the exact optimal regularization strength and L1/L2 mixing ratio are unknown. Our reproduction re-ran the grid search without a fixed random seed, which can yield a different best hyperparameter combination across runs. Given that Elastic Net performance on high-dimensional genomic data is sensitive to small changes in lambda, this difference alone is sufficient to explain the observed gap.

Elastic Net	MAE	MedAE	R^2
Original	12.0	11.0	0.73
Reproduced	14.3	12.4	0.63

Table 3. Performance metrics for Elastic Net regression in the original and reproduced experiments.

To further verify the reproduction, we reconstructed the predicted versus true age scatter plot from the original paper in Figure 5. We see a similar general trend as the paper's results, with comparable dispersion around the identity line across the full age range.

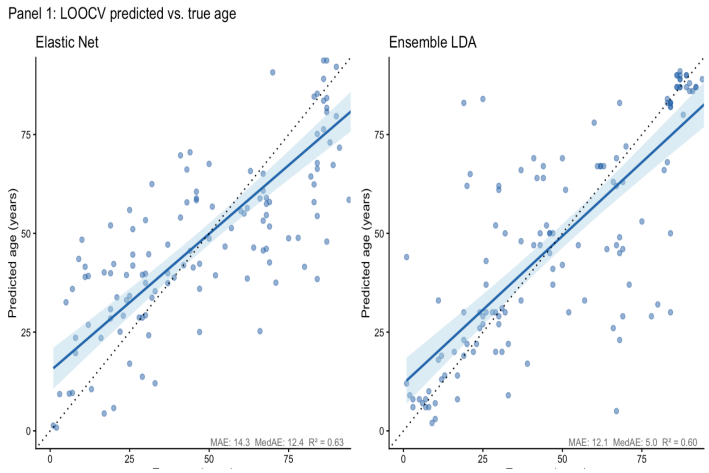


Figure 5. Comparison of Predicted and True Ages for Elastic Net and Ensemble LDA Models.

These results validate the Ensemble LDA as the best and reliable classification model of aging, as it is able to accurately interpret its predictions on HGPS samples as meaningful indicators of accelerated biological age rather than model noise.

New Models

The Neural Network achieved an MAE of 15.52 years, a MedAE of 12.91 years, and an R^2 of 0.37, indicating that the first two principal

components capture approximately 37% of the variance in biological age. This places its error within the range of 7-19 years reported across models in the original study. Based on the error rates, our model is still competing with some of the better original models, but the extreme reduction of active genes into just two principal components may be why our model, in a sense, may be underperforming. Furthermore, the model is struggling with biological heteroscedasticity, where the rate of molecular aging varies significantly between individuals of the same chronological age. This is likely due to the neural network being trained on the average aging trend, thus disregarding external factors like lifestyle or environmental changes.

The XGBoost model achieved an MAE of 18.33 years, a MedAE of 7.93 years, and an R^2 of 0.304. While it produced a higher error rate than the neural network, it explained less variance overall, likely because tree-based decision boundaries isolate specific clusters more tightly in ways that do not generalize as well to continuous age prediction. The similarity in performance between both new models suggests a potential performance floor, given the information retained in just two principal components. Despite experimenting with deeper architectures, the similar results of both of our high-level models indicate that further gains would likely require either additional principal components or richer input features.

To further analyze the results of the extended analysis, as well as continue techniques from the original analysis, as shown in Figure 6, the clustering of points around the blue regression line indicates that while both models successfully capture aging, the significant spread suggests they struggle with precision for individual samples. The models are most accurate in the middle-age range (25-75 years), where the points cluster most tightly, but they tend to struggle at the extremes, often under-predicting the age of older individuals.

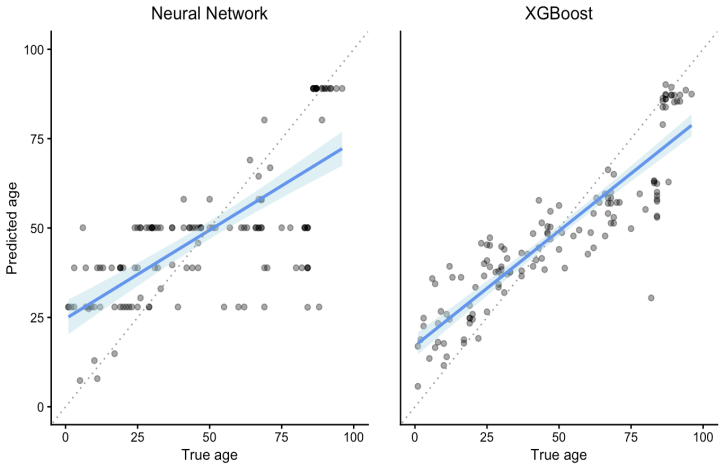


Figure 6. Comparison of Predicted and True Ages for Neural Network and XGBoost Models.

DISCUSSION

Interpretation

The reproduction partially succeeds. The Ensemble LDA and Elastic Net models capture genuine age-related signals in the gene expression data, and the big finding was that the Ensemble LDA outperforms both new and replicated approaches. Our Ensemble LDA MedAE is within 1 year of the paper's, which is encouraging given the implementation differences. Overall, the results also highlight a clear PCA-based tradeoff: reducing the feature space to two principal components improves interpretability and reduces overfitting but limits the models' ability to capture finer biological variation, which likely contributes to the performance ceiling observed across the newer models. This constraint is especially evident in the new models, which show similar levels of performance despite increased model complexity, suggesting that gains are no longer driven by model architecture but by feature representation.

Limitations

This study is limited by the decision to reduce the feature space to only two principal components, which, while improving interpretability, may restrict the models' ability to capture additional biological variation in gene expression. The relatively small sample size, particularly for the Hutchinson-Gilford progeria syndrome group,

further constrains the robustness of subgroup comparisons and limits the generalizability of findings. Lastly, differences in preprocessing choices, software environments, and hyperparameters definitely prevent us from getting to the original results.

Future Research

Future studies can look at expanding the feature representation beyond two principal components to assess whether additional features improve predictive accuracy. Incorporating multi-tissue RNA-seq datasets and even expanding the dataset to include more patient samples, or even testing with other diseases outside of HGPS, like Werner or Cockayne syndrome, could help validate our findings. Further work could also explore nonlinear dimensionality reduction techniques such as autoencoders or t-SNE to retain more complex genomic data. Finally, integrating RNA-seq data with other molecular biomarkers like DNA methylation could improve biological age estimation and provide a more holistic view of aging processes.

AI Disclosure

As a general disclosure, generative AI was used in parts of this assignment. As researchers, we are most comfortable using R to conduct data analysis and modeling; however, the original study was trained in a Python environment. Primarily, its use was focused on debugging during the initial data wrangling stage, as there were a number of issues with matching the dataset to the metadata, handling missing samples, and correcting how the data was structured in matrix form. These fixes ultimately ended up being relatively straightforward to resolve, but, connecting to the prior section, AI can become a crutch moving forward in developing larger and more complex models for genomic analyses.

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